AdaBoost — Short Notes

- **AdaBoost (Adaptive Boosting):** Combines multiple **weak learners** to create a strong classifier.
- **Weak Learner:** A simple model slightly better than random guessing; commonly a **Decision Stump** (tree with one split).
- **Sequential Learning:** Learners are trained one after another.
- **Focus on Errors:** Misclassified samples get **higher weights**, so the next learner focuses more on them.
- **Learner Weight (α):** Better-performing learners get higher weight in the final prediction.
- **Final Prediction:** Combines all learners using their weights.

$$Final = sign\left(\sum_{i=1}^{n} \alpha_i \, h_i(x)\right)$$

Remember:
AdaBoost = Train → Find errors → Increase their weights → Train again → Combine learners.

AdaBoost Algorithm — Short Notes

1. Initialize Weights

For **n data points**, assign equal weight:

$$w_i = \frac{1}{n}$$

2. Train Weak Learner

Train a **Decision Stump (max depth = 1)** and choose the split with the highest **Information Gain**.

3. Calculate Error

$$E = \sum \text{weights of misclassified samples}$$

4. Calculate Learner Weight

$$\alpha = \frac{1}{2}\ln\left(\frac{1-E}{E}\right)$$

- Lower error → **higher α**
- Higher error → **lower α**

5. Update Data Weights

- ❌ Misclassified → **increase weight**
- ✅ Correctly classified → **decrease weight**
- Normalize weights so:

$$\sum w_i = 1$$

6. Resampling

Create a new training dataset according to the updated weights.

Higher-weight samples → higher chance of being selected.

7. Repeat

Train multiple weak learners sequentially, each focusing more on previous mistakes.

8. Final Prediction

Combine all weak learners using their **α weights**:

$$Final = sign\left(\sum_{i=1}^{n} \alpha_i \, h_i(x)\right)$$

⭐ **Remember**

Equal weights → Stump → Calculate error → Calculate α → Update weights → Resample → Repeat → Weighted prediction